Bayesian Graphical Modeling

Network Comparisons

PsyNets Winterschool 2026

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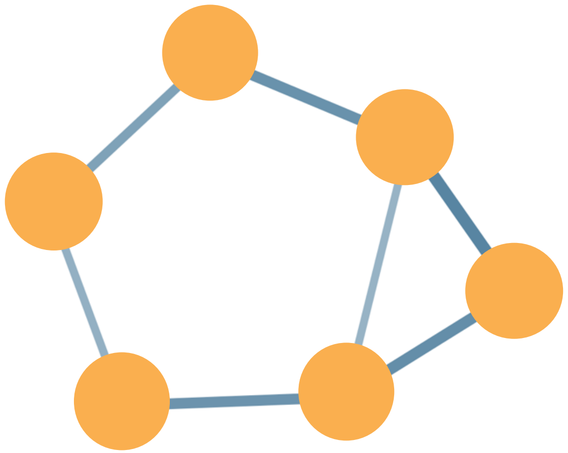
bayesiangraphicalmodeling.com

UvA

Setup Today



Estimating & Testing

Network Comparison

Clustering

Morning Session

Afternoon Session

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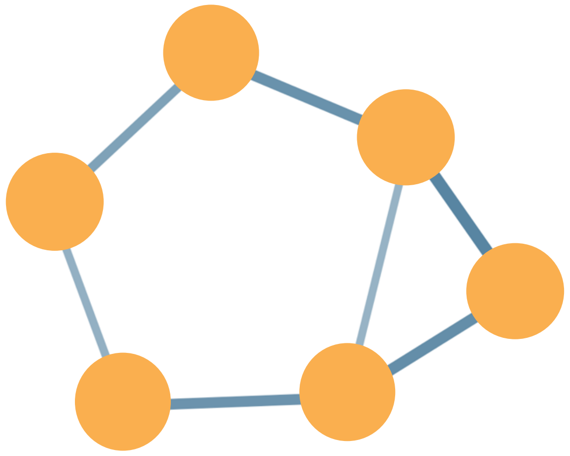

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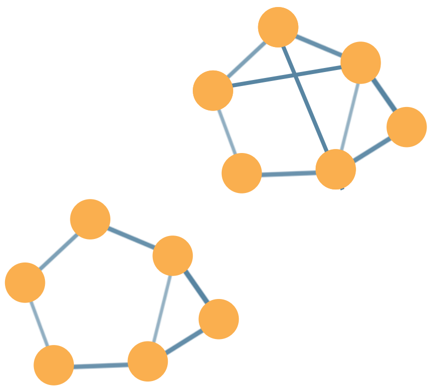

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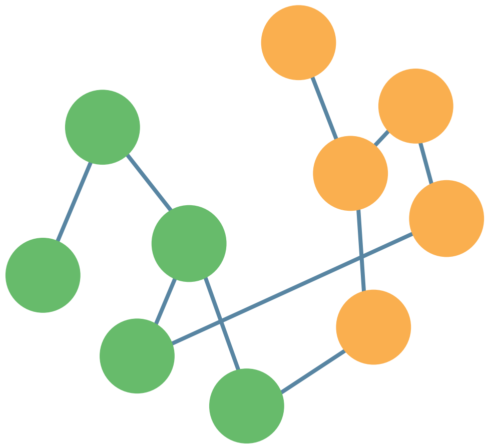

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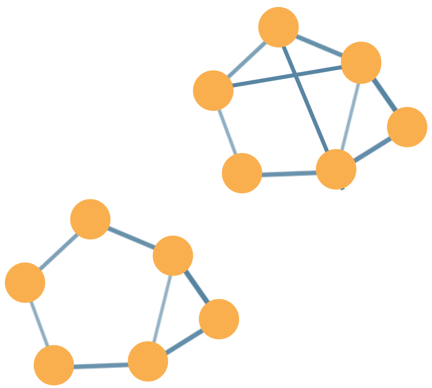

Network Comparison

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Session setup

Part I: Theory

Part II: Practical

Part III: Project

image: Flaticon.com

Session setup

Part I: Theory

image: Flaticon.com

Why Compare Networks?

Why Compare Networks?

1. **Assess network variance or invariance**
    - Do network structures (response behavior) differ between groups?
    - Which relations are stable (invariant) across groups?
    - Where do key differences emerge?

This is a **psychometric approach**, very similar to questions about *differential item functioning* or *measurement invariance* in IRT/SEM frameworks.

Why Compare Networks?

1. **Assess network variance or invariance**

2. **Reproducibility**
    - Do network results reproduce in different samples?
    - Do network results generalize to different populations?

A central motivation for developing Bayesian approaches to group comparisons is the concern about *reproducibility* in network psychometrics.

This is a **methodological approach**, aiming to distinguish signal from noise.

Why Compare Networks?

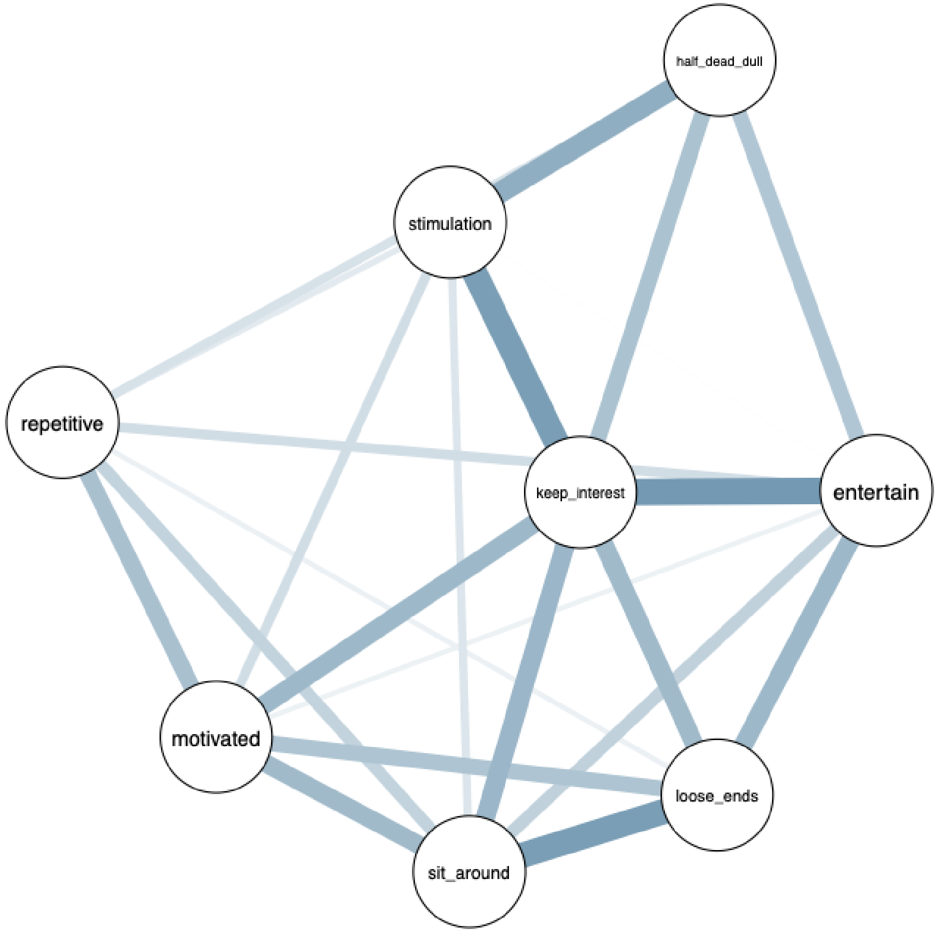
1. **Assess network variance or invariance**

2. **Reproducibility**

3. **Asses theoretical predictions**
   - Do empirical networks match theoretical expectations?
     For example, expected subgroup differences (e.g., involvement)

Network models are not only descriptive; they can also be used to formalize and test theories. This offers a **substantive perspective** to network comparison.

Boredom Proneness Network



Nodes are short boredom proneness scale items (7-point Likert-scale).

Edges are partial associations.

Goal is to determine conditional dependence relationships.

```
library(bgms)
?Boredom
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Martarelli, et al. (2023). A trait-based network perspective on the validation of the French short boredom proneness scale. *European Journal of Psychological Assessment*, *39*(6), 390-399.

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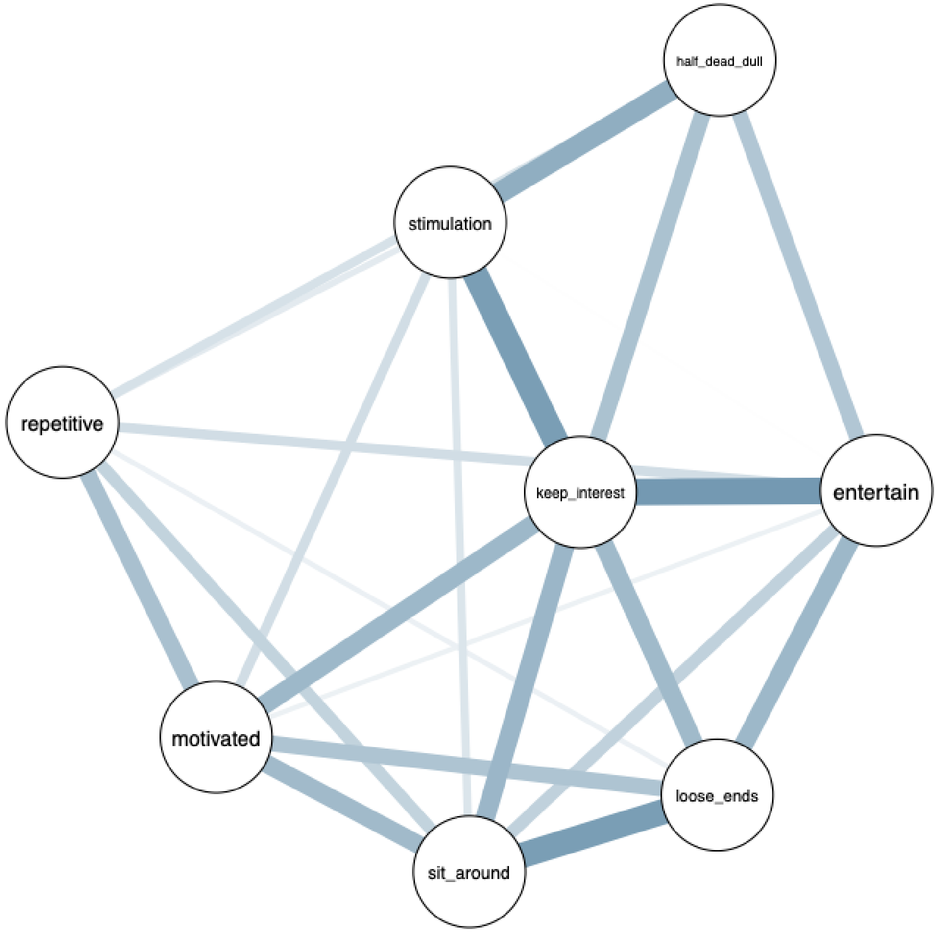


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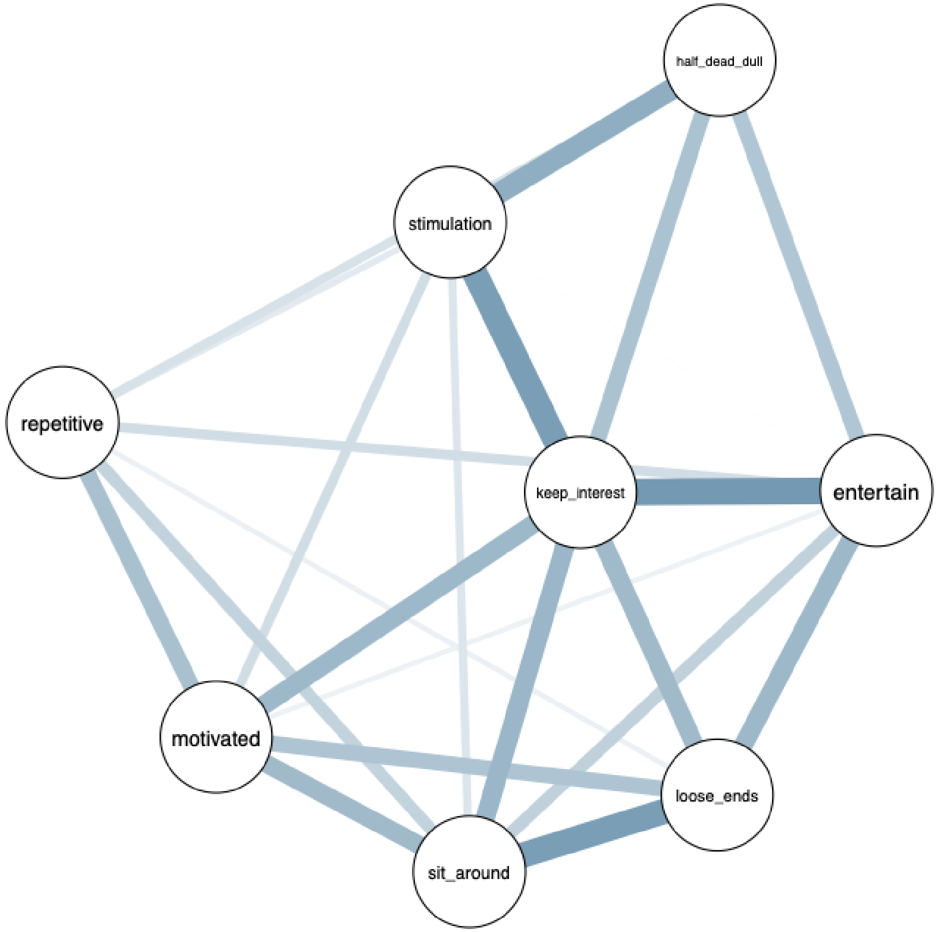


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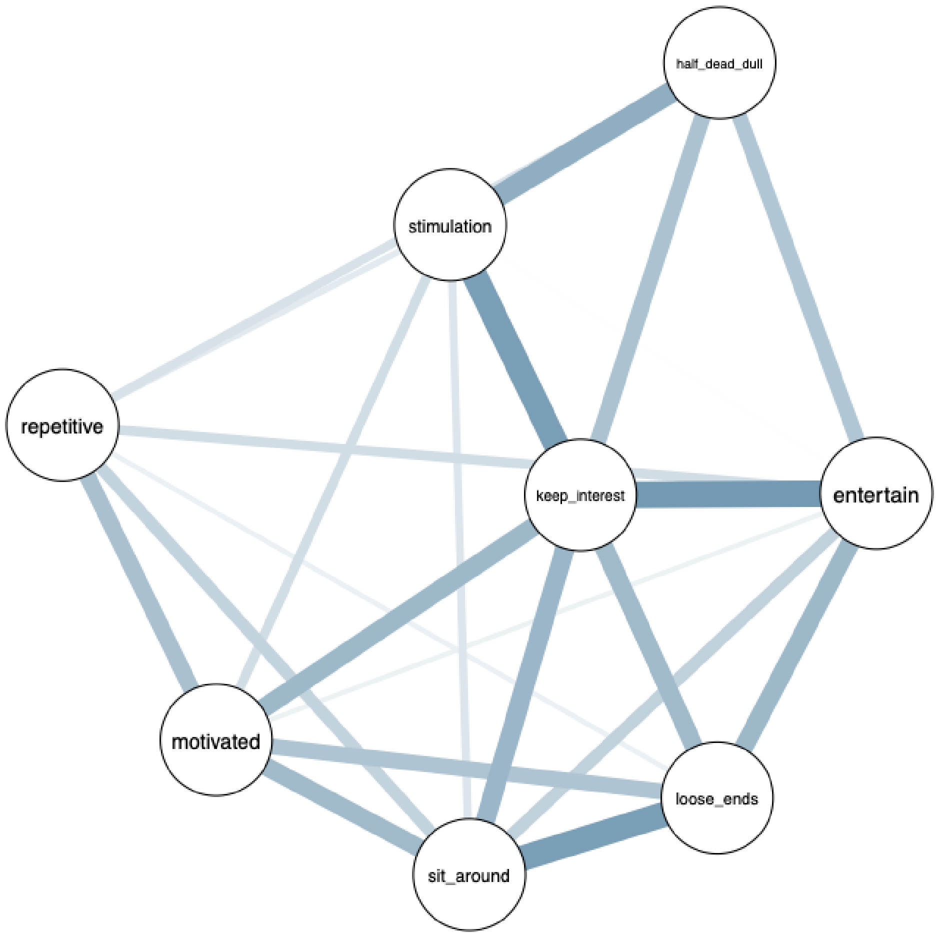


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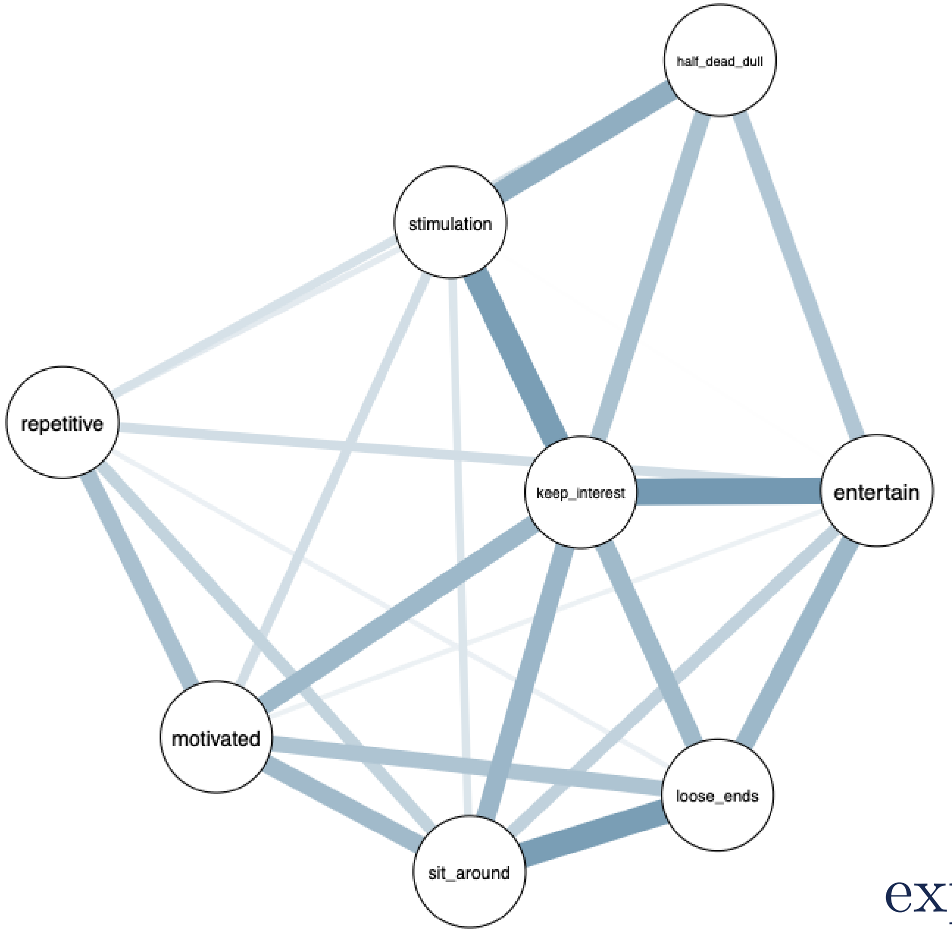
The Ordinal Markov Random Field



$$\frac{\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}{\sum_{\mathbf{x}}\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}$$

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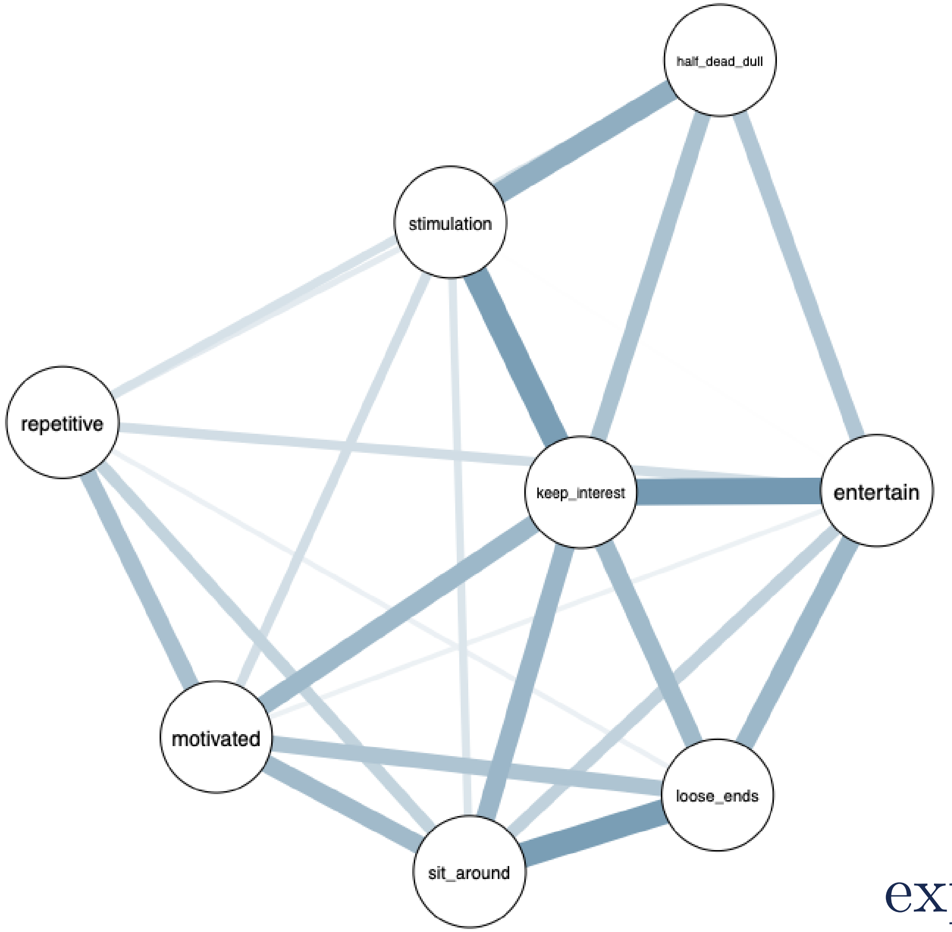

Nodes are binary or ordinal variables.

$$\frac{\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}{\sum_{\mathbf{x}}\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}$$

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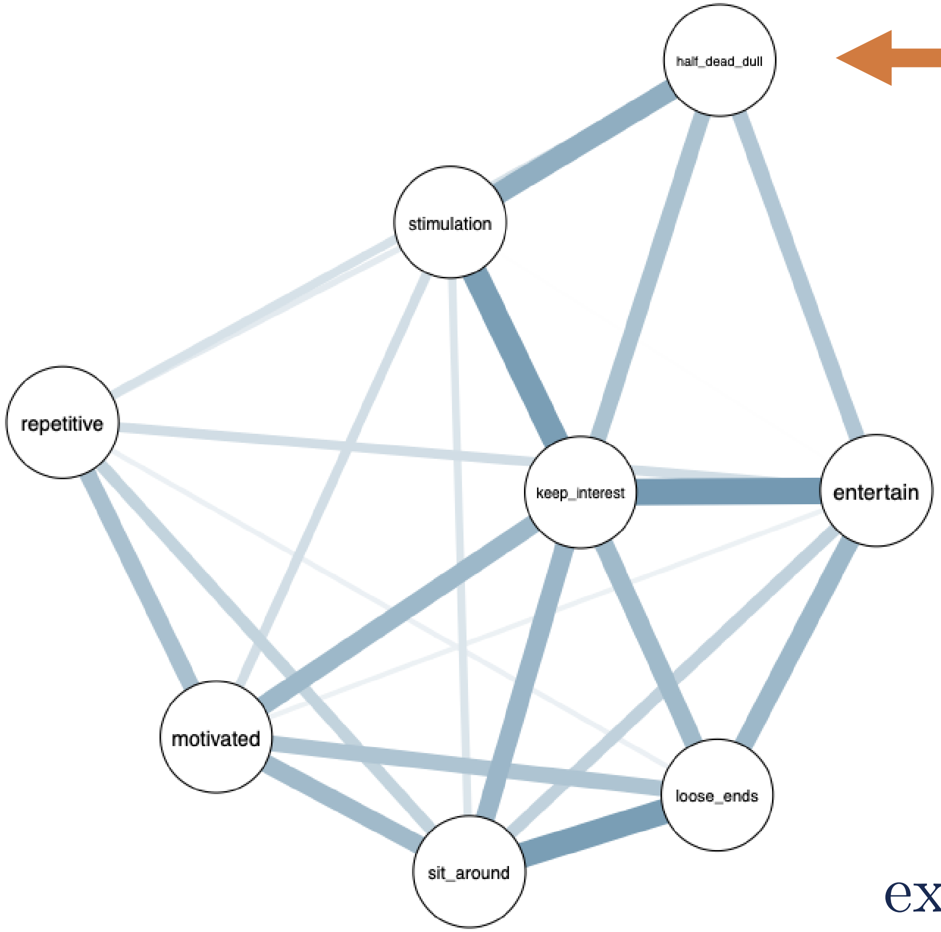
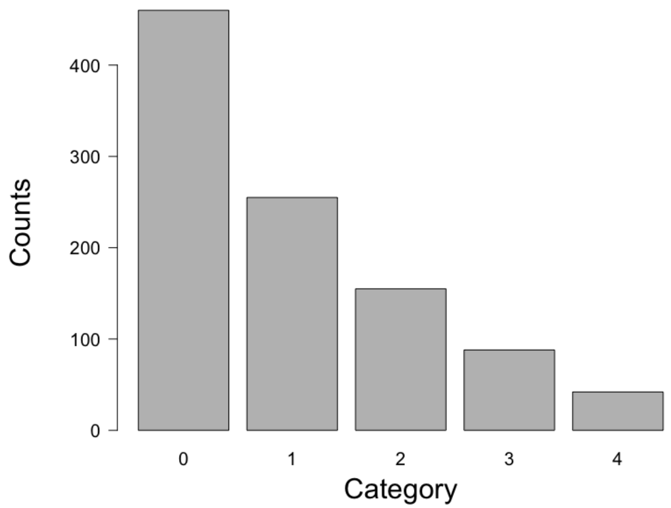


$\mu_{i.}$

$\mu_{j.}$



$$\frac{\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\boxed{\mu_{ic}}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}{\sum_{\mathbf{x}}\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}$$

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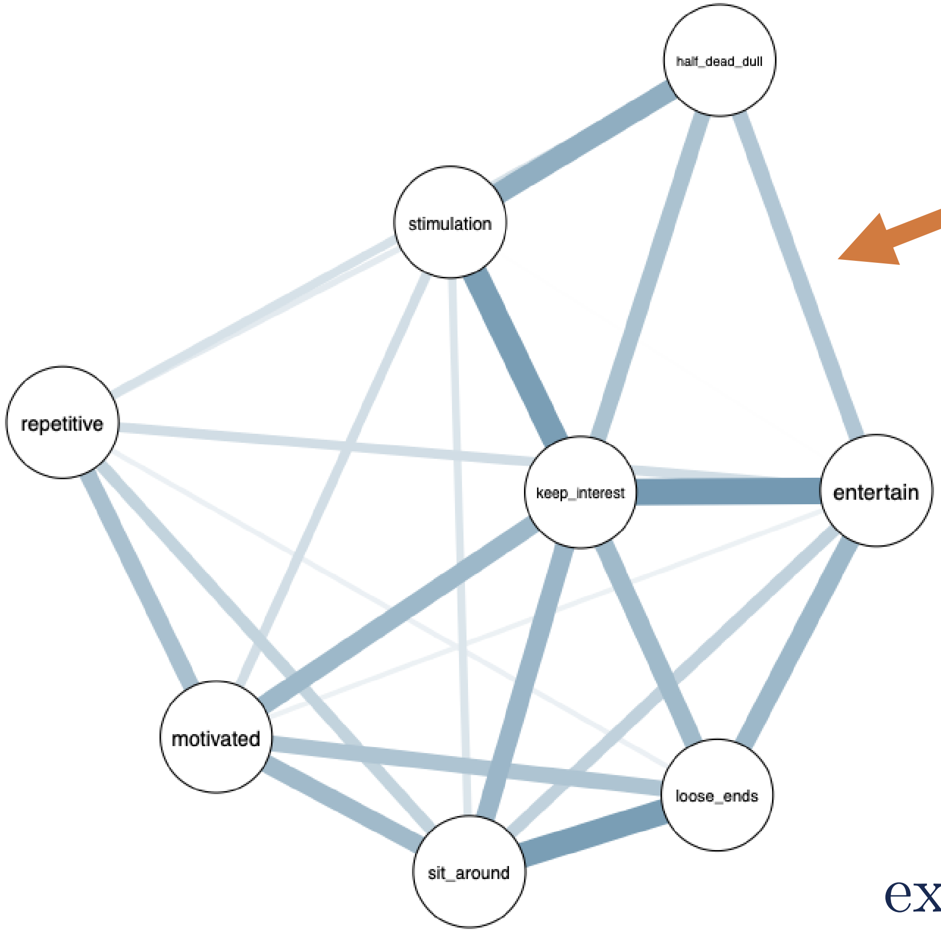
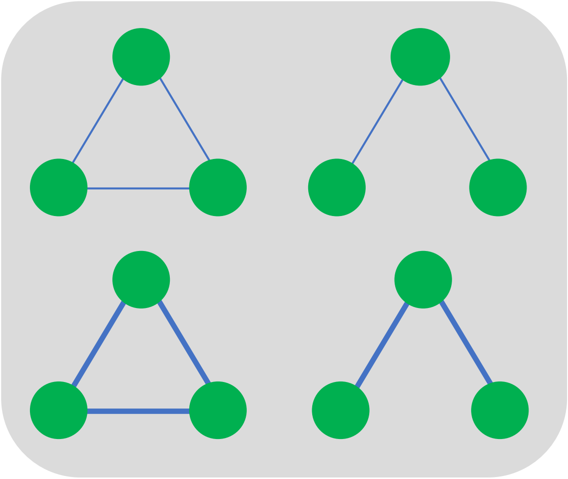


σ_{ij}



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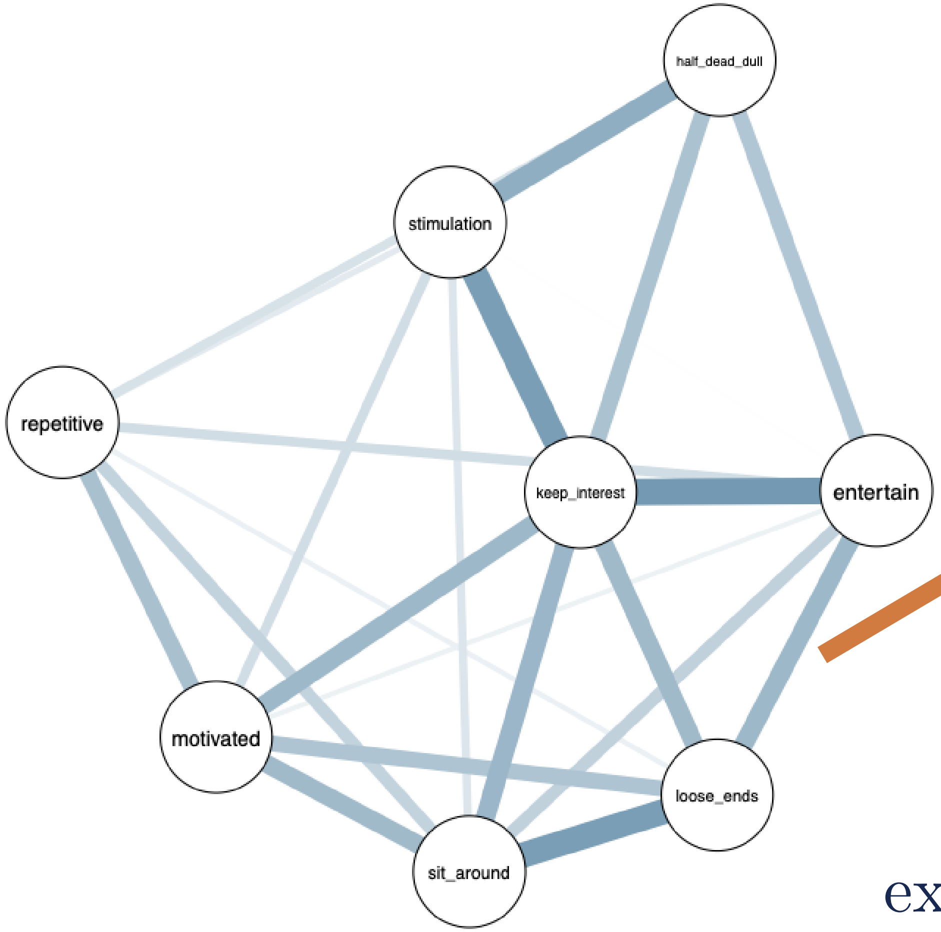

SPECIFIC OVERVIEW

Relation	Estimate	Posterior Incl. Prob.	Inclusion BF	Category
entertain-loose_ends	0.116	1.000	Inf	included
repetitive-loose_ends	0.016	0.353	0.546	inconclusive
stimulation-loose_ends	0.079	1.000	Inf	included

$$\frac{\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}{\sum_{\mathbf{x}}\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}$$

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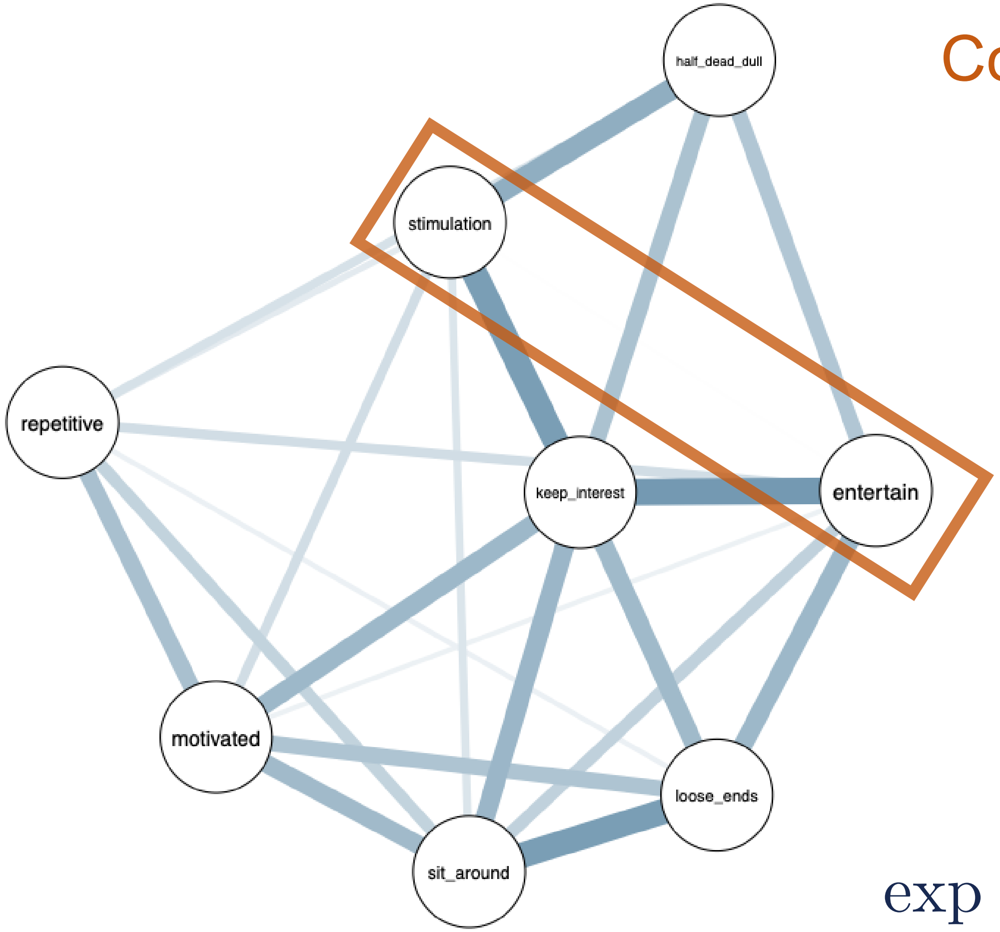
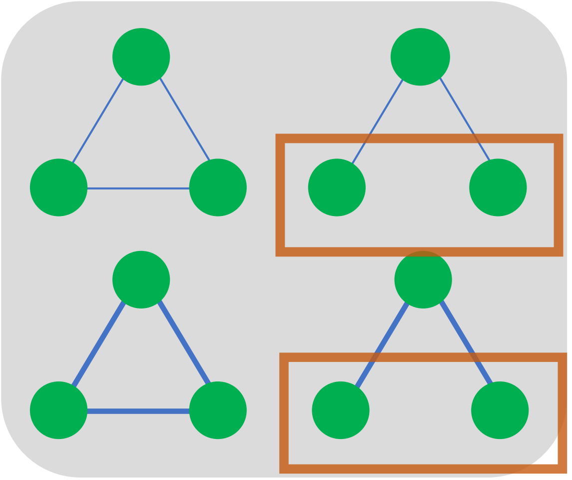


Conditional independence?



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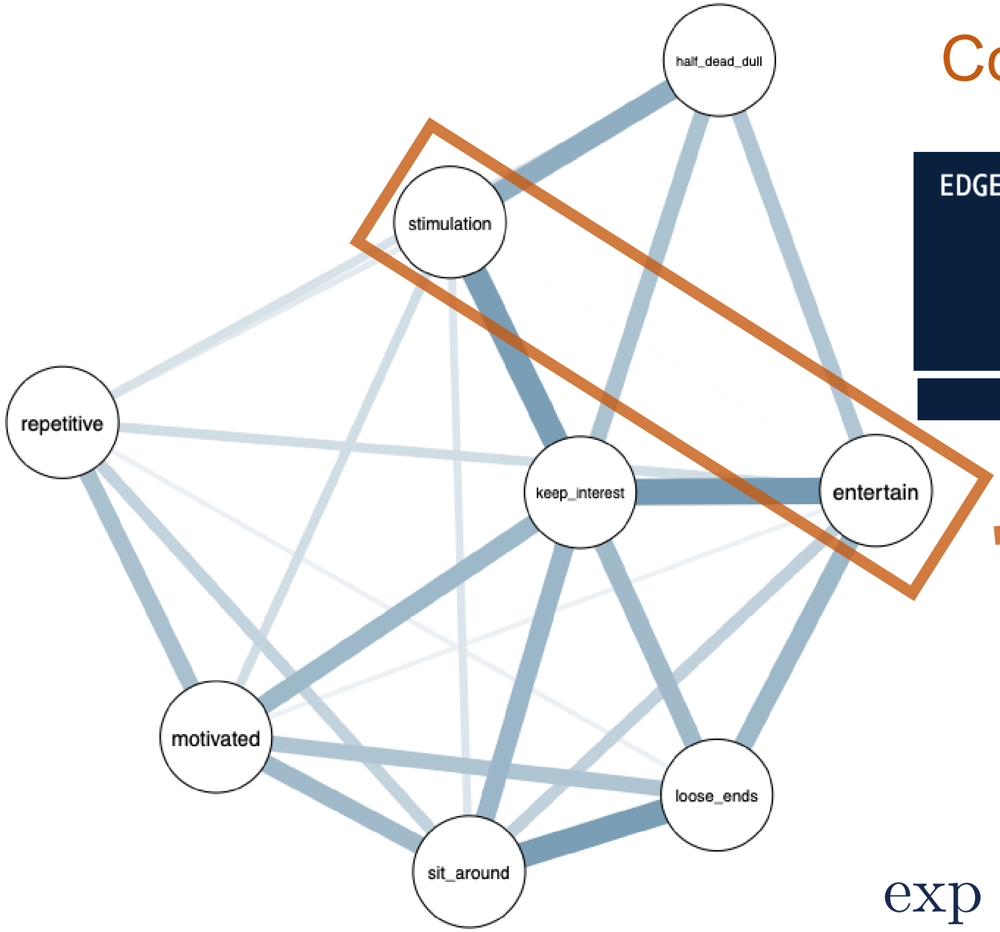

Conditional independence?

EDGE SPECIFIC OVERVIEW

Relation	Estimate	Posterior Incl. Prob.	Inclusion BF	Category
entertain-loose_ends	0.116	1.000	Inf	included
repetitive-loose_ends	0.016	0.353	0.546	inconclusive
stimulation-loose_ends	0.079	1.000	Inf	included
stimulation-entertain	0.001	0.041	0.043	excluded

$$\frac{\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}{\sum_{\mathbf{x}}\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i}\mathcal{I}(x_i=c)\,\mu_{ic}+2\sum_{i>j}x_ix_j\sigma_{ij}\right)}$$

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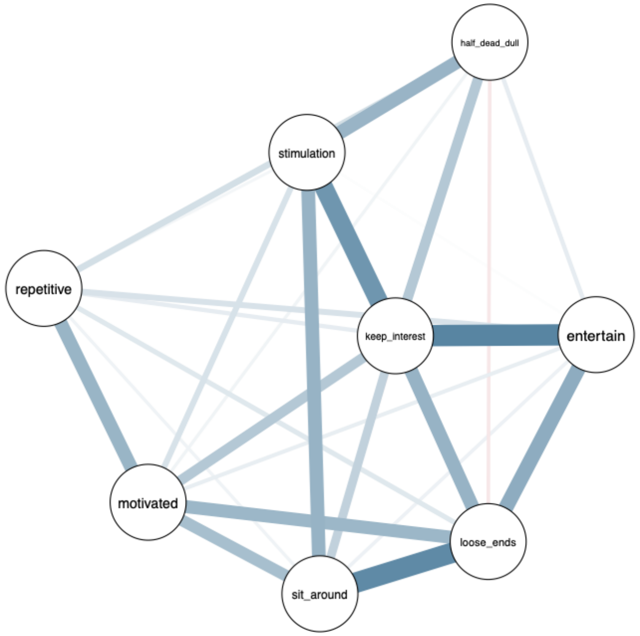
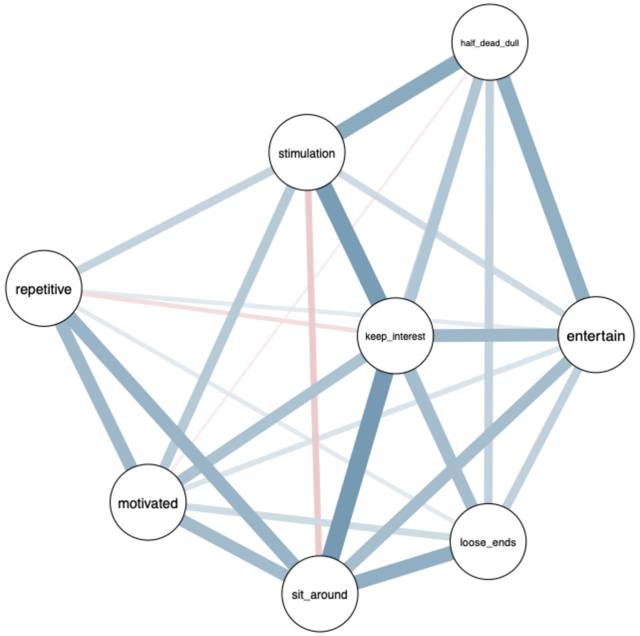
French and English-Speaking Samples



French (n = 496)



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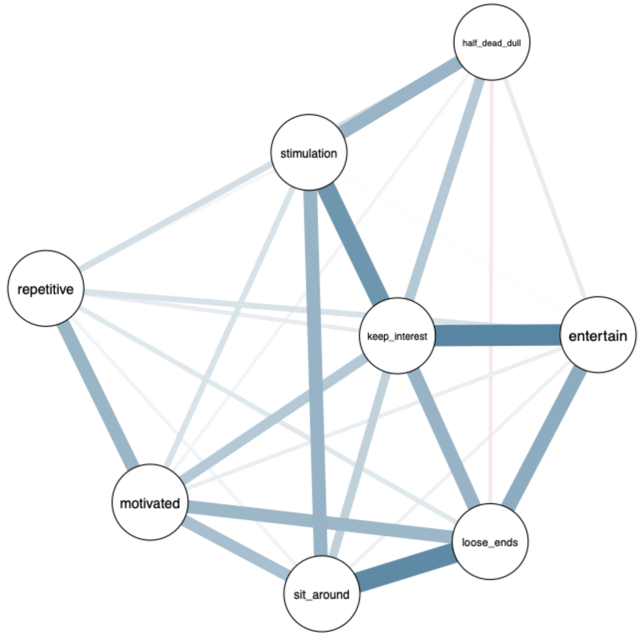

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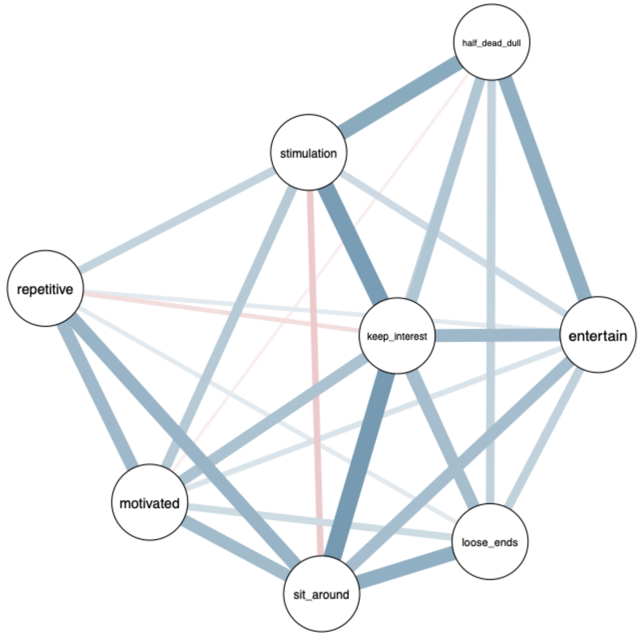

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How could we assess sample differences?

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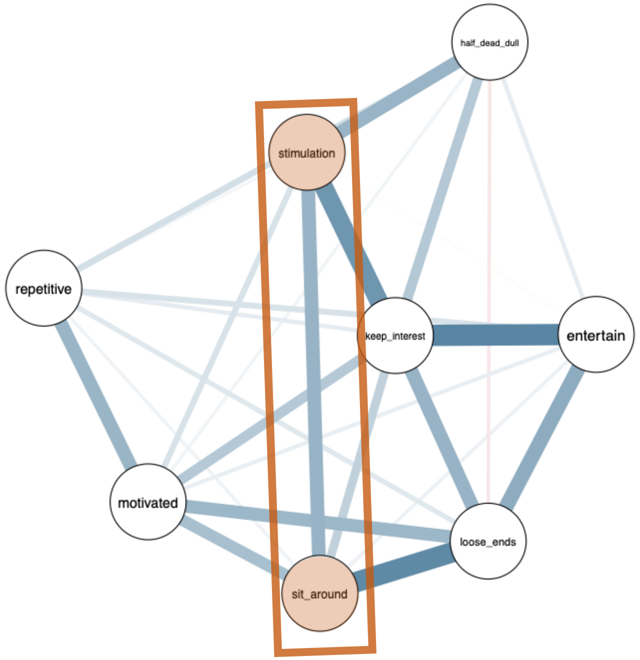

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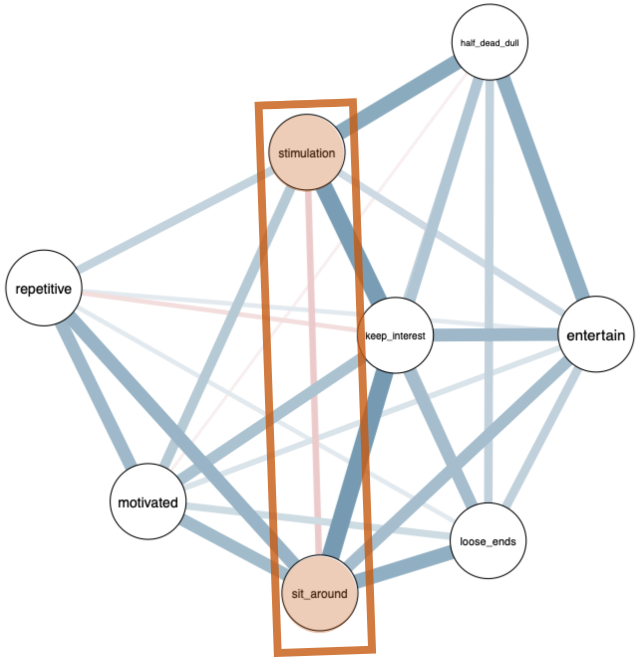

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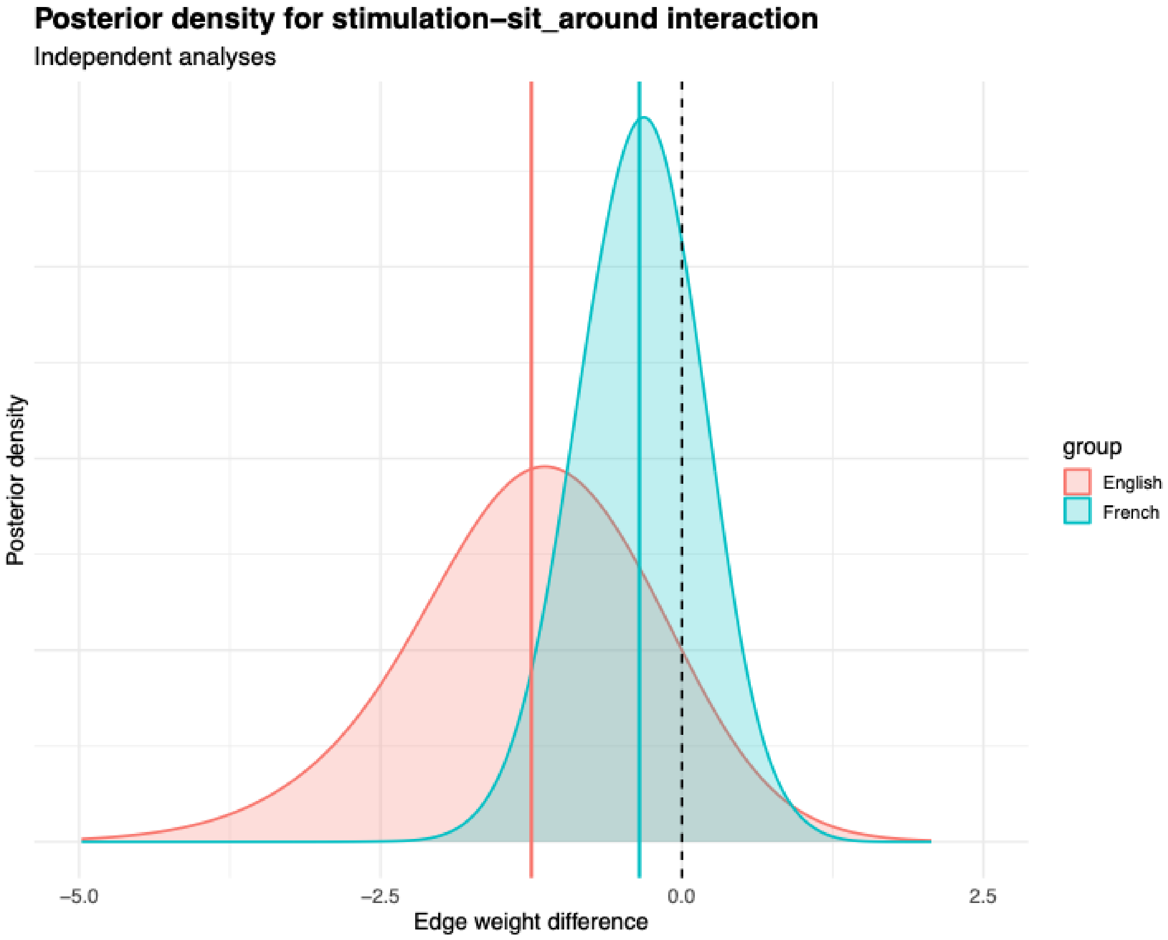

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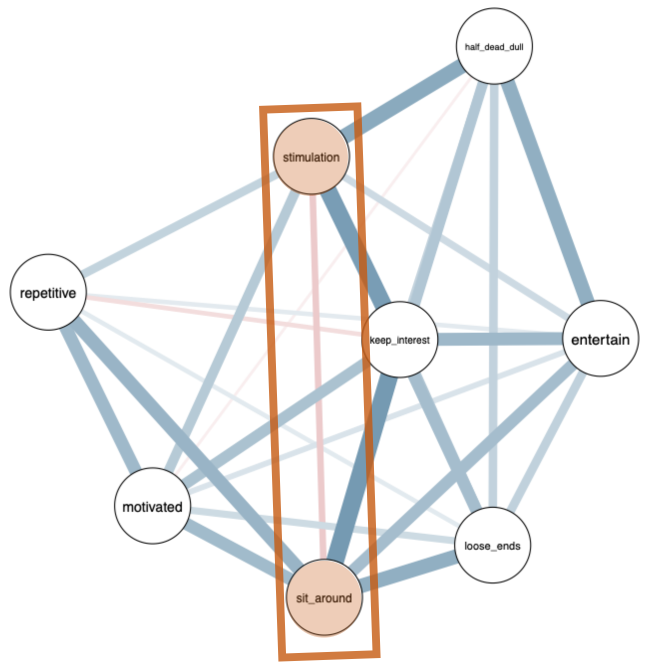

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$$\sigma_{ij} = \begin{cases} \phi_{ij} - \frac{1}{2}\delta_{ij} & \text{for group 1} \\ \phi_{ij} + \frac{1}{2}\delta_{ij} & \text{for group 2} \end{cases}$$

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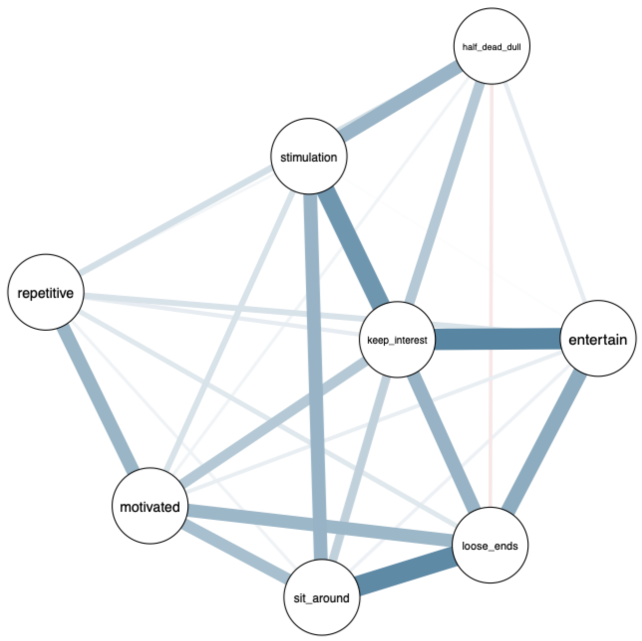
Model the Difference in Edge Weight



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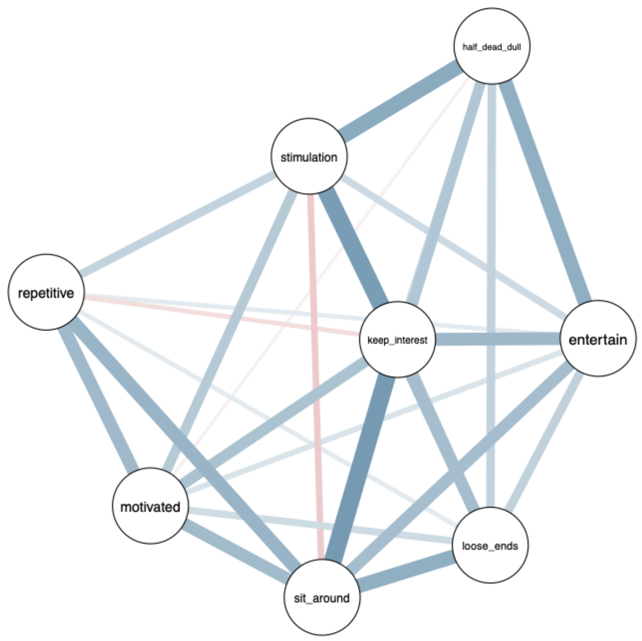

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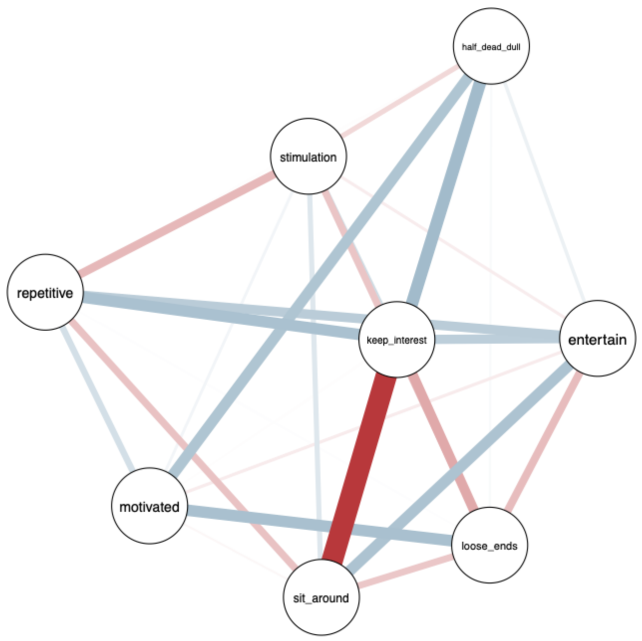

$$\delta = \sigma_{\text{fr}} - \sigma_{\text{en}}$$

Marsman, Waldorp, Sekulovski, & Haslbeck (2025). Bayes factor tests for group differences in ordinal and binary graphical models. *Psychometrika*, *90*, 1809-1842.

Test the Difference in Edge Weight

Rather than testing whether an **edge weight** should be included or excluded (i.e., conditional independence), we now test if an **edge difference** should be included or excluded (i.e., edge invariance).

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$$\mathcal{H}_0\text{: Edge is equivalent in the two groups} \iff \delta_{ij} = 0$$

$$\mathcal{H}_1\text{: Edge is different in the two groups} \iff \delta_{ij} \neq 0$$

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$$\text{BF}_{10} = \frac{p(\mathcal{H}_1 \mid \text{data})}{p(\mathcal{H}_0 \mid \text{data})} \Bigg/ \frac{p(\mathcal{H}_1)}{p(\mathcal{H}_0)}$$

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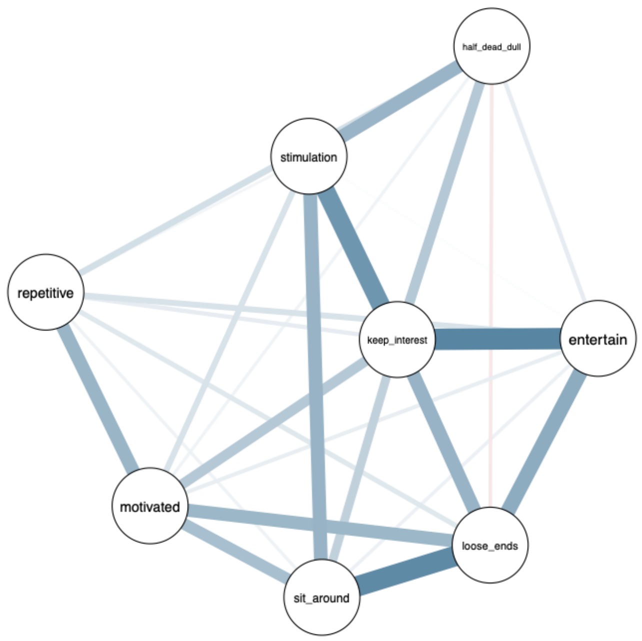

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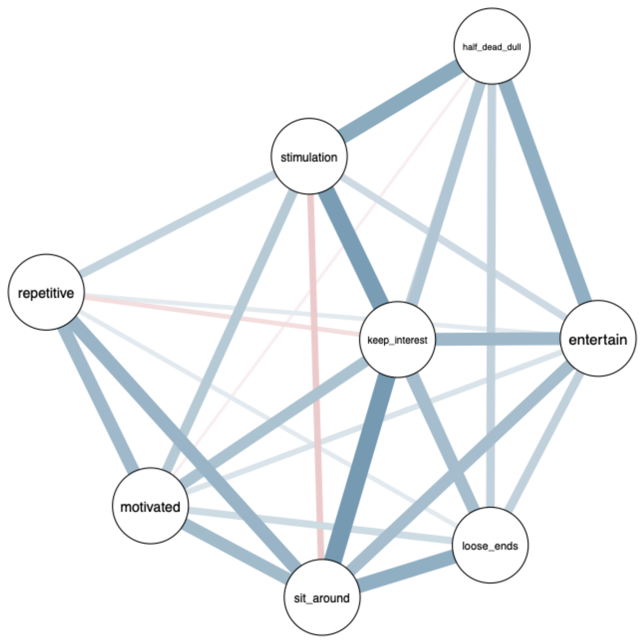

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Classic vs Bayesian Network Comparison

Classic Approaches

Network Comparison Test (NCT)

- Permutation-based test
- Returns a *p*-value for rejecting equality

Fused Graphical Lasso (FGL)

- Jointly estimates networks with penalization
- Encourages similarity across groups
- No inferential evidence for difference or invariance

Limitations

Asymmetry of evidence

Can reject equality, but not support it

Multiplicity problems

Edge-wise testing inflates false positives or requires harsh corrections

Visual fallacy

Missing edges are often interpreted as “no difference”

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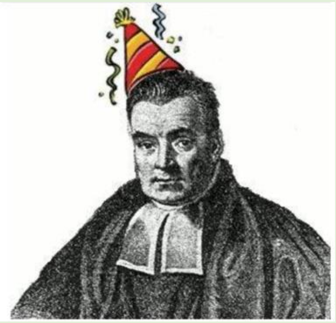

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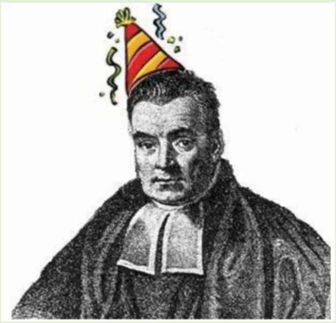


Advantages

Bayes factors

Quantify evidence *for differences* and *for invariance*

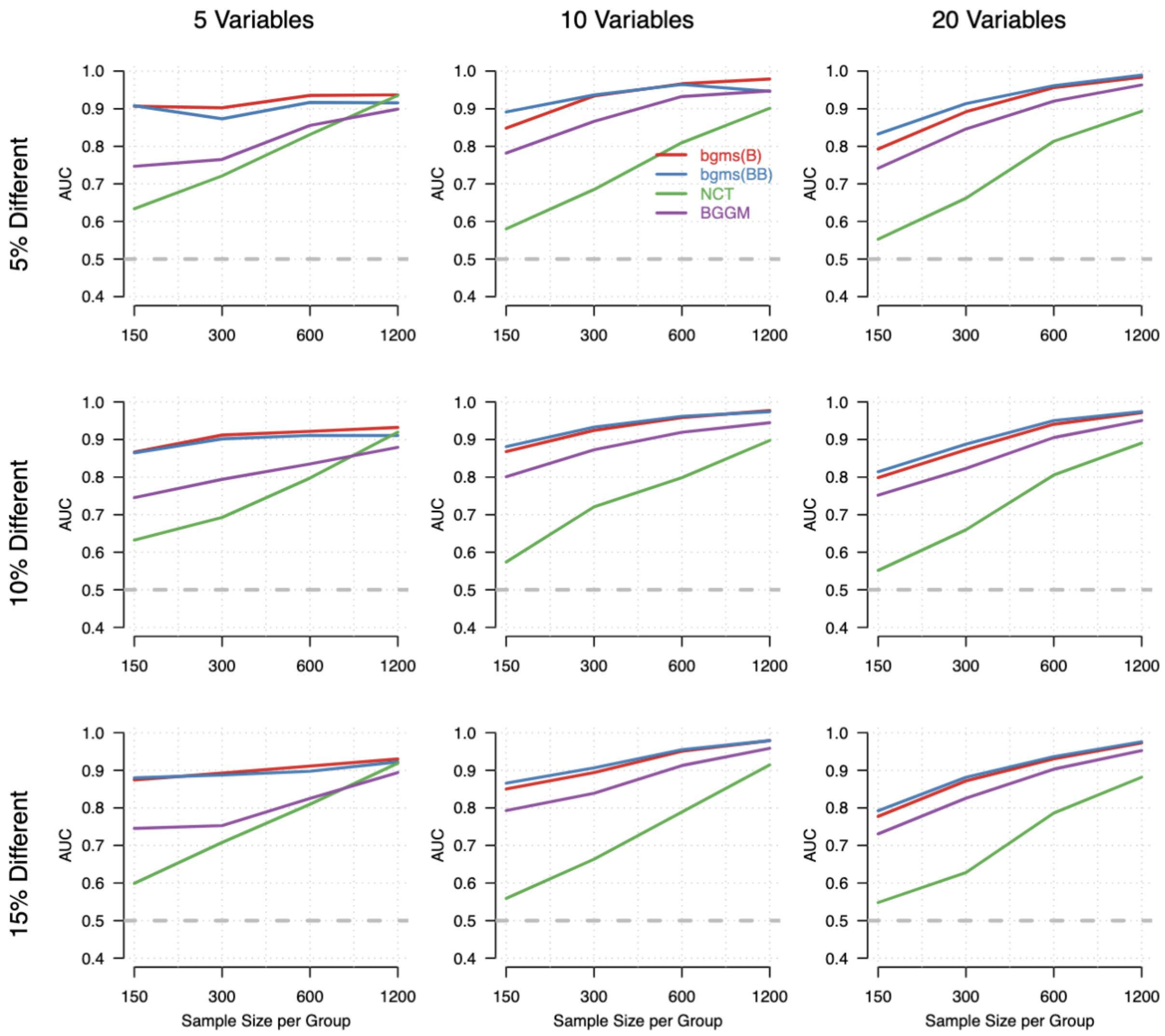
Model averaging

Can automatically correct for multiplicity

Posterior distributions and evidence plots

Make uncertainty explicit

Simulation Evidence



Area under the ROC curve

- Threshold-free summary of detection accuracy
- 0.5 = chance; higher = better

Design:

- Variables: 5, 10, 20
- Edge differences: 5–15%
- Sample size: 150–1200 per group

Results:

- Bayesian MRF methods perform best overall
- Strong gains in small samples and low signal
- NCT competitive with large samples & few variables

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Session setup

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image: Flaticon.com